

Artefact 1: MRP Data Analysis and Planning View

Data Analyst Internship - ST Engineering

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Confidentiality note: The sample data in this artefact is created for demonstration. It does not contain actual company materials, quantities, screenshots or business rules.

1. Description of the deliverable

This artefact shows how I organised dated material demand and supply records into a clearer planning view. It includes a mock source extract, the transformation steps, and a simple horizontal output that can be reviewed by date.

My contribution: I reviewed the source field structure, clarified the meaning of the MRP elements, cleaned and grouped the records, and designed the date-based output.

How it will be used: The processed structure can support planning review and can later be used as an input for a dashboard after the business rules and source data are validated.

2. Mock source data

Material	Required date	MRP element	Reference	Quantity
DEMO-FG-100	15 Aug 2026	Independent requirement	DEMO-REQ-01	-5
DEMO-FG-100	01 Aug 2026	Warehouse stock	DEMO-STOCK	+2
DEMO-FG-100	10 Aug 2026	Planned order	DEMO-PLAN-01	+3
DEMO-COMP-20	08 Aug 2026	Dependent requirement	DEMO-DEP-01	-10
DEMO-COMP-20	06 Aug 2026	Purchase order	DEMO-PO-01	+6

3. Work process

- Checked what each date, MRP element and quantity sign represents.
- Standardised the field names and kept the reference number for traceability.
- Separated confirmed information from assumptions that still required clarification.
- Sorted the movements by date and calculated the projected balance.
- Presented the result in a horizontal format so shortages and coverage dates are easier to see.

4. Sample output

Material	Opening	06 Aug	08 Aug	10 Aug	15 Aug	Status
DEMO-FG-100	2	-	-	+3 planned	-5 demand	Covered by required date
DEMO-COMP-20	0	+6 PO	-10 demand	-	-	Shortage of 4

From the mock data, DEMO-COMP-20 is still short by four units after the purchase order. In actual work, any action must be based on complete data and approved planning rules.

5. Technical skills and competency level

Technical skill / competency	Level	Evidence and area for improvement
Data Analytics	Level 2 - Apply	I cleaned, classified and interpreted structured records. I still need guidance when validating manufacturing-specific rules.
Data Visualisation	Level 2 - Apply	I created a clearer date-based view. More user testing is needed to confirm that the layout is practical.
Business Needs Analysis	Level 2 - Apply with guidance	I clarified field meanings and expected outputs instead of making assumptions.

6. Generic skills used

Critical Core Skill	Task / action	Result
Sense Making	Compared dates, field meanings and MRP element types.	Confirmed what could be used and recorded what was still unclear.
Digital Fluency	Used spreadsheet and structured-data techniques to prepare the output.	Produced a reusable sample structure.
Communication	Asked for clarification when field definitions were unclear.	Reduced the chance of presenting the wrong interpretation.
Problem Solving	Investigated missing fields and incomplete calculation inputs.	Recorded the limitations instead of overstating the result.

7. Corrective action and reflection

Some field meanings and calculation inputs were incomplete. I corrected this by checking the definitions, separating assumptions from confirmed rules, and avoiding unsupported calculations. The next step is to validate the logic using approved data and obtain feedback from the intended users.

This task taught me that data preparation is not only about changing the layout. I also need to understand the business meaning of each field and clearly state the limitations before the output can be used for decision-making.